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FACULTY OF MATHEMATICS AND COMPUTER
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Computer Science in English

BACHELOR THESIS

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2023

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Abstract

The software application presented in this thesis is called **CV Builder** and it is a secure and free tool for whoever wishes to create their own digital resume. The design is clean, professional looking and the user experience while using the app should be a pleasant one due to the easy to use interface and the light design.

It was build using modern and fast technologies that performed impeccably and allowed for learning opportunities. The structure is modular and components can be re-utilized, thus, future changes can be easily made. It presents itself with a layered architecture that has 3 tiers: the presentation tier, or user interface, the application tier and the data tier. This type of architecture offered a solid base for the successful development up until this point and is capable to support further development.

Furthermore, it presents the development of a complete resume generator application designed to incorporate the best features of existing tools such as Europass, Resume.com, and Canva while addressing their shortcomings. A critical review of these platforms highlighted limitations such as limited customization options, restrictive free tier capabilities, and overwhelming design variety. In response, my developed application provides an user-friendly interaction, substantial customization, and completely free services.

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Chapter 1

Introduction

1.1 Background Study

Study results show that the onset of the pandemic created an immediate increase in job applications, and this practice still endured into the post-onset period.[11] So it's quite plausible that the number of remote job interviews has also increased considerably.

Depending on the job type the onset and post-onset applications were substantially greater for work-from-home jobs (which followed a negatively accelerated curve) compared to face-to-face jobs.[11]

The demand for remote work in the aftermath of the pandemic has drastically increased. With the job-seeking pool being incredibly large, it has become beneficial for both the employer and the job-seeker to use online tools in order to conduct the job application process.[7]

It has reached the level where in certain fields, the first part or even the entire hiring process takes place online. Therefore, the steps that must be taken until employment are also in the online environment. Everything starts of course with looking for a job, but after that, the next step is applying for that job by sending a CV.

The CV serves as the initial impression an individual makes on potential employers, and a well-organized CV is more likely to distinguish them from other applicants and create a positive impact.

A strong CV increases the chances of being noticed by recruiters and hiring managers by highlighting skills and accomplishments. A query conducted by "Career Builder" concluded that 61% of employers believe soft skills are just as important as hard skills.[2] Consequently, this can lead to increased prospects for career progression and higher salaries.

Nevertheless, creating an impactful CV can be challenging and time-consuming, particularly for those lacking prior experience in this area.

1.2 Motivation

From my personal experience, I know that creating CV can be an excruciating and exhausting task, especially for the inexperienced. As such, an up-to-date app that specialises in building CVs is to be desired - not only for the inexperienced, but also for convenience's sake. Supporting the end user with tips and overall guidance is also an important factor that has to be taken into account, qualities which are uncommon in today's Resume creators.

Furthermore, given the ever-increasing demand for remote work, I believe that developing a digital tool of this nature can greatly benefit users by helping them create professional-looking resumes that accurately portray their skills and experience. The app should have a user-friendly interface and use current technologies, such as AI, to provide recommendations for improving the CV based on the user's education, skills, and work experience.

1.3 Description

In today's world, it is almost unthinkable to apply for a job without submitting a CV. As mentioned by the research[7], job applications are usually done online, competition is increasing as it only necessary to submit a digital copy of the resume. Therefore, it is imperative to stand out from other applicants from the very first moment.

The aim of this app is to assist the user thought the creation process of a Resume. A clear, simple resume that uses logical layouts is desired, as according to a recent study made by "The Ladders", they have a much better chance of being noticed by recruiters.[8]

To use the app, users simply need an email address which they will use to create an account. They can select an area of interest and be presented with different templates. After that, they can save their personal data, so they will need less time if they want to create multiple resumes.

Chapter 2

Related Apps

2.1 Europass

Europass[3] is a European Union initiative aimed at supporting the mobility of citizens across Europe by providing standardized tools for documenting skills, qualifications, and experiences. The Europass portfolio includes five documents that help individuals present their skills and qualifications in a clear, easy-to-understand format, but we will focus only on Europass CV [4].

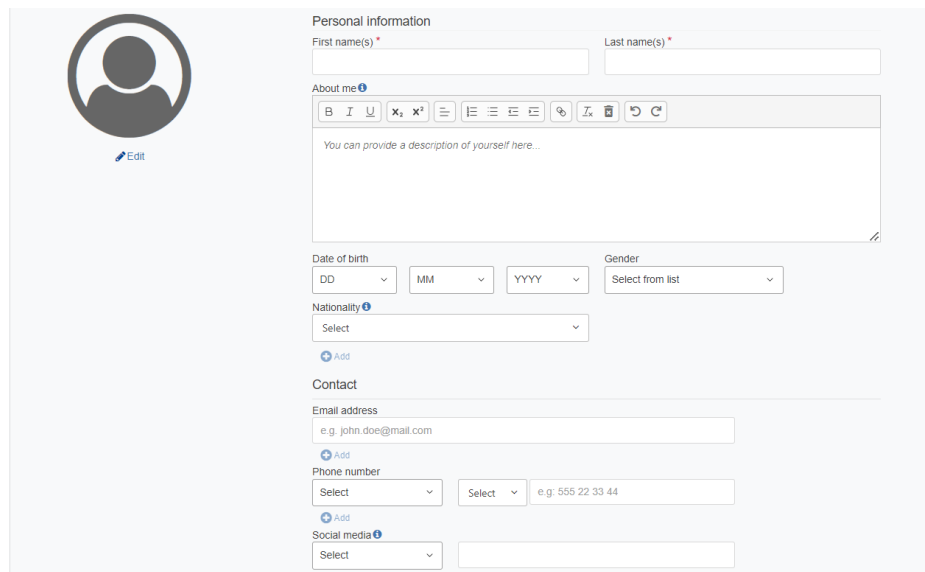
The image shows a screenshot of the Europass CV form. On the left, there is a circular profile picture placeholder with a blue 'Edit' link below it. The main form area is titled 'Personal Information' and contains several fields: 'First name(s) *' and 'Last name(s) *' are text input fields; 'About me' is a rich text editor with a toolbar (bold, italic, underline, bulleted list, numbered list, link, unlink, image, video, table) and a placeholder text 'You can provide a description of yourself here...'; 'Date of birth' consists of three dropdown menus for 'DD', 'MM', and 'YYYY'; 'Gender' is a dropdown menu with 'Select from list'; 'Nationality' is a dropdown menu with 'Select' and an 'Add' link; 'Contact' section includes 'Email address' (text input with placeholder 'e.g. john.doe@mail.com' and an 'Add' link), 'Phone number' (three dropdown menus for country code, area code, and number with placeholder 'e.g. 555 22 33 44' and an 'Add' link), and 'Social media' (dropdown menu with 'Select' and an 'Add' link).

Figure 2.1: Europass UI[3]

Advantages:

- **Standardized format:** The Europass CV provides a standardized template for presenting your skills and qualifications, which can help make your application more clear and easily understandable to employers or other organizations.
- **European recognition:** The Europass CV is recognized throughout the European Union, so it can be helpful if you're applying for jobs or educational opportunities across borders.

- **Comprehensive:** The Europass CV includes sections for personal information, education and training, work experience, skills, and other relevant information, which can help ensure that you're presenting a complete and comprehensive picture of your qualifications.
- **User-friendly:** The Europass CV is designed to be easy to use and fill out, so you don't need specialized knowledge or skills to create one.
- **Multilingual:** The Europass CV is available in many different languages, which can be helpful if you're applying for jobs or educational opportunities in a language other than your native language.

Disadvantages:

- **Lack of customization:** The standardized format of the Europass CV may not allow for as much customization or personalization as other types of CVs, which may make it difficult to highlight your unique skills or experiences.
- **Limited space:** The Europass CV has a set format and limited space, so it may be challenging to fit all of your relevant qualifications and experiences into the available sections.
- **Not universally recognized:** While the Europass CV is recognized throughout the European Union, it may not be as well-known or widely used in other parts of the world, which could limit its usefulness if you're applying for jobs or educational opportunities outside of Europe.
- **Overemphasis on formal education:** The Europass CV places a strong emphasis on formal education and training, which may not accurately reflect an individual's skills or experiences in other areas.
- **Potential language barriers:** While the Europass CV is available in many different languages, using a CV template that isn't in your native language could potentially introduce errors or make it more difficult to express yourself clearly.

Overall, using the Europass CV can help you present your skills and qualifications in a clear, standardized way that is recognized throughout Europe, which can potentially increase your chances of success in your job or educational pursuits. While the Europass CV can be an useful tool, it may not be the best choice for everyone, particularly if you need more flexibility or customization in your CV.

2.2 Resume.com

Resume.com is a website that offers a range of services related to job searching and career development. The site's main focus is to provide users with a tool for creating professional-looking resumes quickly and easily. Resume.com's resume builder allows users to choose from a variety of customizable templates and input their work experience, skills, and education. The site also offers cover letter templates and a job search feature that allows users to search for job openings and apply directly through the site.

Resume.com also provides various resources to help job seekers improve their job search and career development skills. The site features articles and blog posts on topics such as resume writing, interviewing tips, and career advice. Users can also access career assessment tools to help them identify their strengths and weaknesses and determine which careers might be a good fit for them.

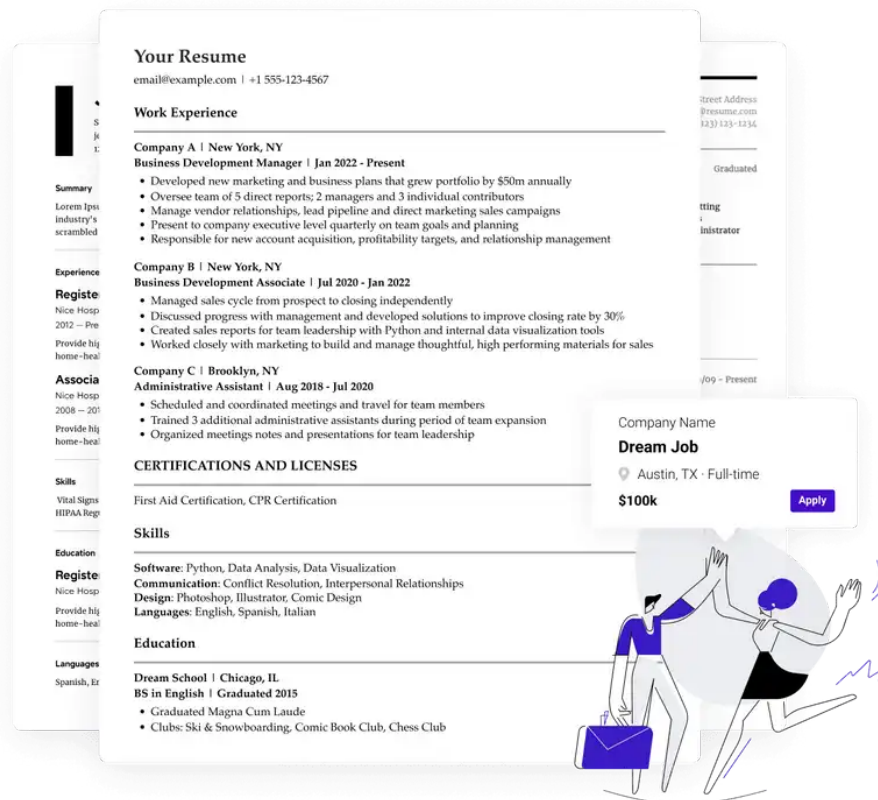


Figure 2.2: Resume First Screen[12]

Advantages:

- **User-friendly resume builder:** The website's resume builder is user-friendly and straightforward, making it easy for anyone to create a professional-looking resume without any prior design experience.
- **Customizable templates:** Resume.com offers a variety of customizable resume templates, which users can adapt to fit their specific needs and preferences.
- **Cover letter templates:** In addition to resume templates, the site offers cover letter templates that users can customize and tailor to the job they are applying for.
- **Job search feature:** Resume.com has a job search feature that allows users to search for job openings by location and keyword. Users can also upload their resume to apply for jobs directly through the site.
- **Career resources:** The site offers a range of career resources, including

articles, blog posts, and career assessment tools, to help users improve their job search and career development skills.

- **Free basic plan:** Resume.com offers a free basic plan, which includes access to the resume builder and a limited number of resume templates. Users can upgrade to a paid plan for more features and benefits.
- **Mobile app:** The site also offers a mobile app, which allows users to create and edit resumes on the go, making it convenient for those who are constantly on the move.

Disadvantages:

- **Limited customization:** While Resume.com offers a range of customizable templates, the level of customization is limited compared to more advanced design software. Users may not be able to fully customize the layout or design elements of their resumes.
- **Basic features on the free plan:** The free plan on Resume.com offers limited access to templates and features, and users may need to upgrade to a paid plan to access more advanced features, such as the ability to download and print resumes in different file formats.
- **No option to import resumes:** Users cannot import their existing resumes into the website, which means they need to start from scratch or manually copy and paste their information into the builder.
- **No guarantee of job placement:** While Resume.com can help users create a professional-looking resume and search for job openings, it does not guarantee job placement. Users will still need to actively apply for jobs and go through the standard job application and interview process.
- **Limited job search options:** While the job search feature on Resume.com can be useful, it may not include all available job openings in a given area. Users may need to supplement their job search with other job boards or networking efforts.

Ultimately, Resume.com can be a useful tool for job seekers who want to create a professional-looking resume, search for job openings, and improve their career development skills. However, there may be some limitations that users should consider before signing up.

2.3 Canva

Canva is a popular online platform that offers a wide range of tools for creating visually appealing documents, including CVs. It provides a user-friendly interface and a standardized format that can effectively showcase your skills, qualifications, and experiences.

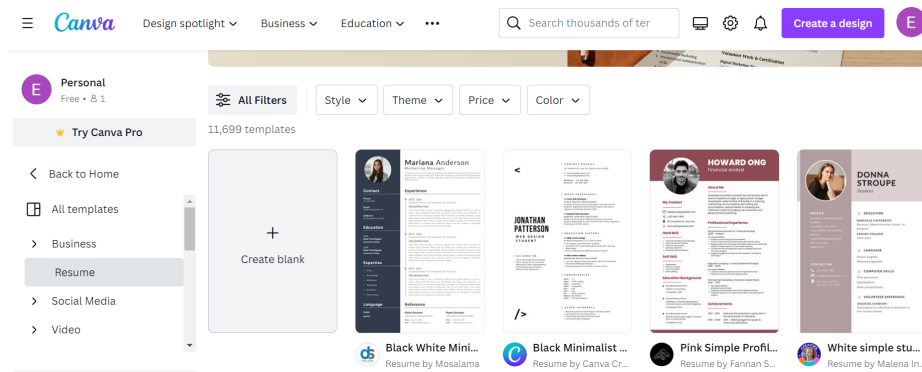


Figure 2.3: Canva UI[1]

Advantages:

- **Standardized format:** Canva offers a variety of templates specifically designed for CVs, ensuring a consistent and professional look that is easy to understand for employers and organizations.
- **Customization options:** While using a standardized format, Canva provides ample customization features that allow you to personalize your CV and highlight your unique skills and experiences.
- **Visual appeal:** With its extensive collection of design elements, Canva enables you to create visually stunning CVs that stand out and make a memorable impression.
- **User-friendly interface:** Canva is designed to be intuitive and easy to use, even for individuals without specialized knowledge or skills in graphic design.
- **Multilingual support:** Canva supports multiple languages, making it suitable for individuals applying for jobs or educational opportunities in different linguistic contexts.
- **Mobile app availability:** Canva's mobile app allows you to create, edit, and access your CV on the go, providing convenience and flexibility to work on your CV anytime and anywhere.

Disadvantages:

- **Limited free features:** While Canva offers a free version, some advanced design elements and templates may require a subscription or payment.
- **Potential overemphasis on design:** As Canva focuses on visual aesthetics, there is a risk of prioritizing design elements over the content and substance of your CV.
- **Online platform dependency:** Creating a CV on Canva requires internet access, and you need to ensure the availability and security of your data while using the platform.
- **Recognition outside design-oriented fields:** While Canva is well-known and widely used in design-related industries, its recognition and suitability may vary in other professional domains.

Overall, Canva, with its mobile app and customizable features, offers individuals a convenient and visually appealing platform for creating CVs. It provides a standardized format that allows for personalization, enabling users to effectively present their qualifications. Whether accessed on a computer or through the mobile app, Canva assists in the creation of professional CVs that have the potential to enhance job applications and educational pursuits. However, it is important to consider the design-focused nature of the platform and evaluate specific requirements when determining if Canva is the suitable choice.

2.4 Critical Analysis

All three apps were a source of inspiration in the process of creating the app. They have good and interesting features, which I tried to reproduce in my application, but also have some downsides.

This application is similar to them, presenting the user with a series of templates from which he can choose what suits him best, depending on the field of activity. Also, to benefit from the application's services, the user must create an account, so that the submitted work is saved inside a portfolio. From there, the user can later edit, delete or download their CV.

In the case of Europass, the application interface is easy to use and the application services are free. Having said that, it does not offer much freedom of customization, the CV models offered are quite limited and the Europass CV model is not universally recognized. This application can improve on these aspects, offering a greater variety of templates from which the user can choose, while still keeping the services free.

Resume.com is quite similar to this CV Builder application in terms of how the application works. In fact, it features even more templates and more complex designs, but all of these are only available as part of the premium plan. Yes, the application also has the option to create a free CV, but the free options are quite limited and therefore not really worth the effort.

As for Canva, the number of templates offered and the level of customization is amazing, with over 11,000 CV models available, a truly impressive number. But, as with Resume.com, all of this comes at a price. To access the full Canva services, you have to pay a premium fee, which can be confusing when you're just starting out and you're not exactly sure what you want. Also, although the level of customization is high, the variety of CVs available can be too much even for the most experienced users.

The application is made with the aim of taking the best aspects of the CV creation tools presented above and trying to improve their weak points. It presents the user with a simple and friendly interface, but at the same time offers a good level of customization while keeping everything free.

Chapter 3

Application Description

3.1 App Requirements

The main expected requirements are for a system user to be able to create an account, log in, choose a domain, then a template for his CV, and finally download a .pdf version of the CV. The web app should guide the user and make its path easier by assisting him.

3.1.1 Use Case Diagram

The Use Case Diagram seen in Figure 3.1 describes how the actors (Users, Administrators) interact with the web interface.

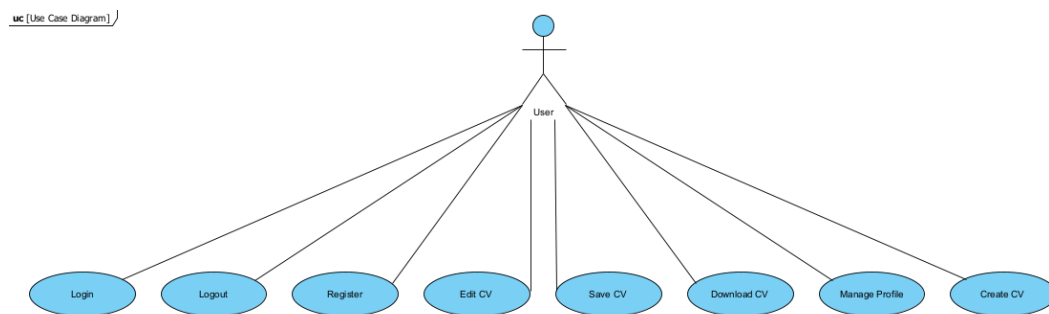


Figure 3.1: Use Case Diagram

3.1.2 Use Case Details

Login:

Actors: User

Description: The user can log in to the CV builder app using their credentials.

Preconditions: The user has a registered account in the CV builder app.

Postconditions: The user is successfully logged into the app and can access the respective functionalities.

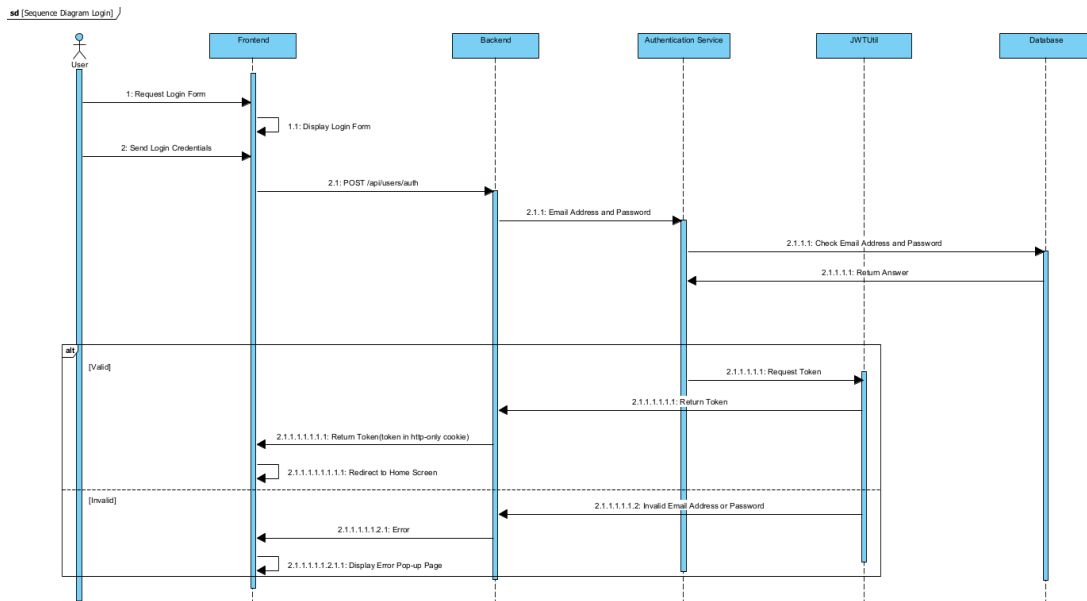


Figure 3.2: Login Sequence Diagram

Register:

Actors: User

Description: The users can register for a new account in the CV builder app by providing their personal information and creating login credentials.

Preconditions: The users have access to the CV builder app registration page.

Postconditions: The user's account is created, and they can log in to the app using their credentials.

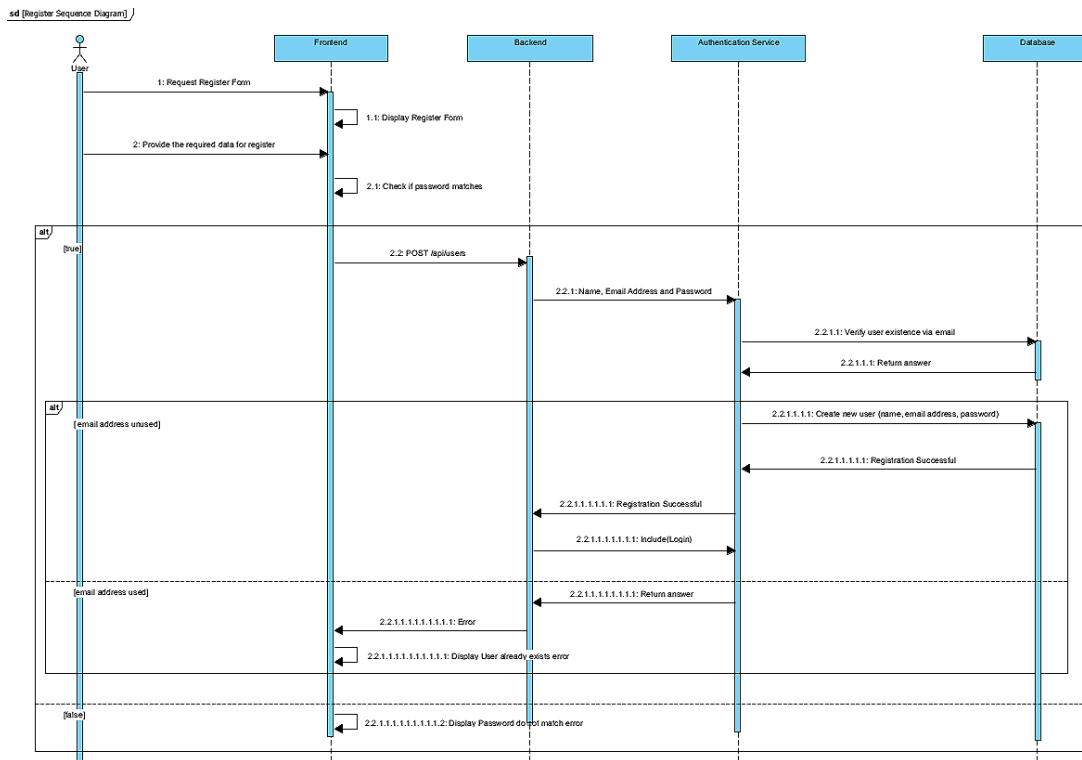


Figure 3.3: Register Sequence Diagram

Create CV:

Actors: User

Description: The user can create a new CV by entering their personal information, work experience, education, skills, and other relevant details.

Preconditions: The user is logged into the CV builder app.

Postconditions: A new CV is created and the information is stored in the app.

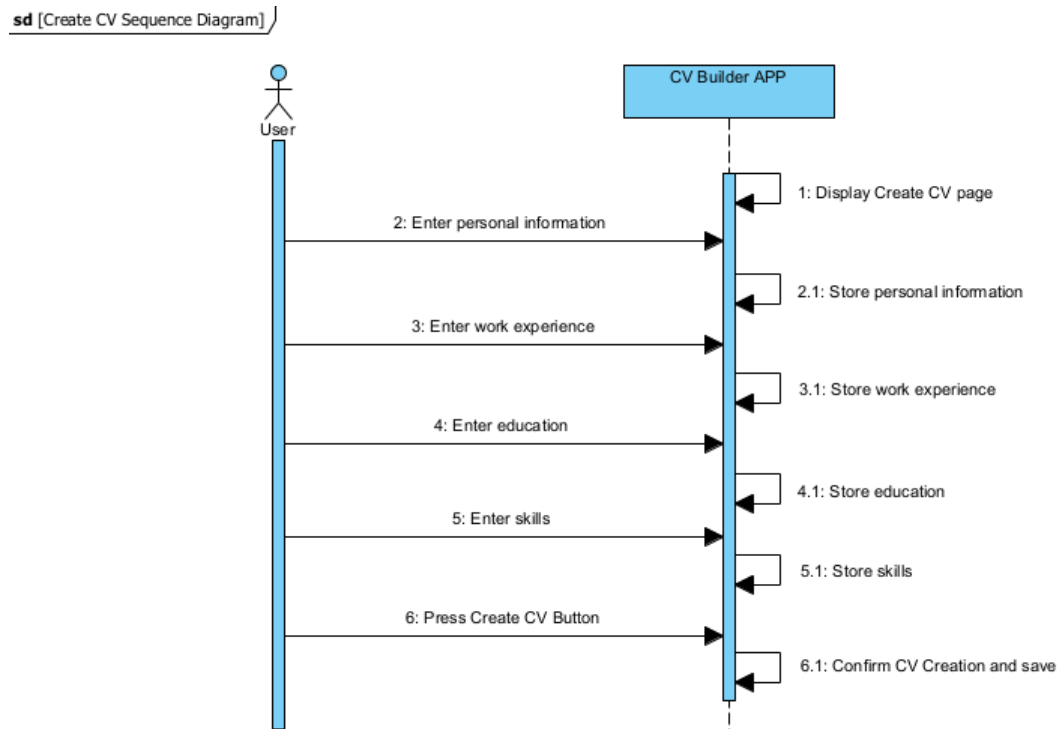


Figure 3.4: Create CV Sequence Diagram

Edit CV:

Actors: User

Description: The user can edit their existing CV by modifying or updating the information entered earlier.

Preconditions: The user has created his CV on the app.

Postconditions: The CV is updated with the new information.

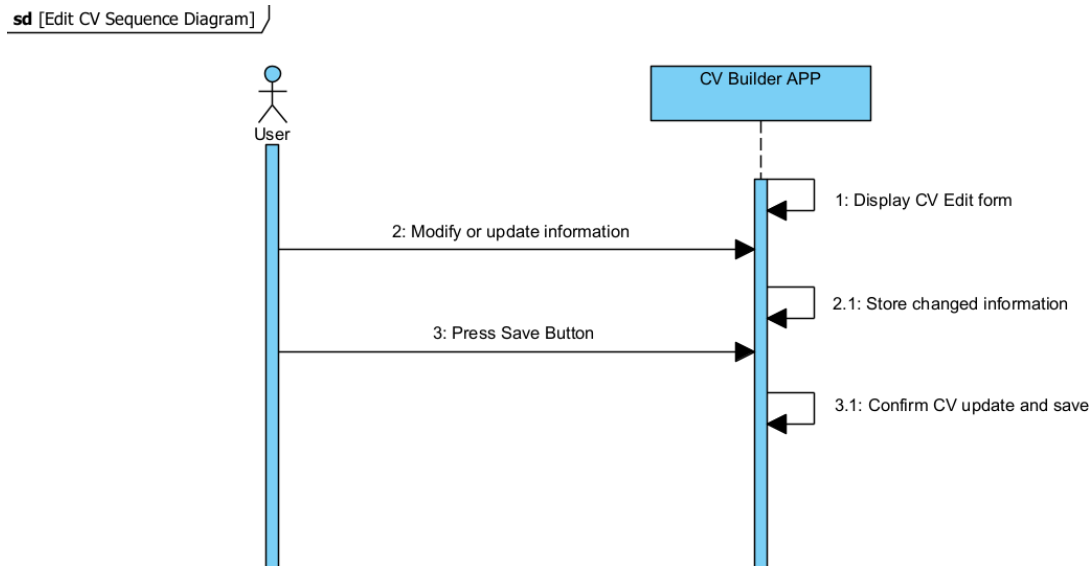


Figure 3.5: Edit CV Sequence Diagram

Download CV:

Actors: User

Description: The user can generate a pdf version of their CV for easy distribution or submission.

Preconditions: The user has a saved CV in the app.

Postconditions: The CV is printed or saved as a printable file.

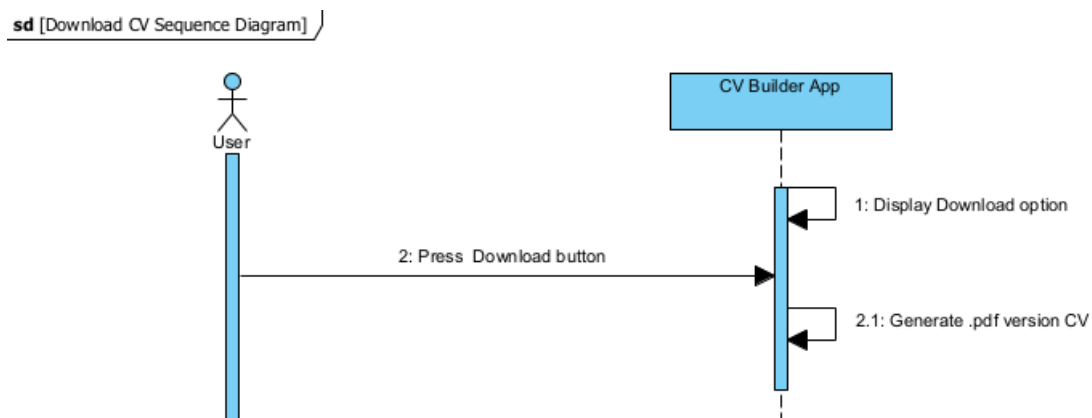


Figure 3.6: Download CV Sequence Diagram

Manage Profile:

Actors: User

Description: The user can view and edit their profile information

Preconditions: The user is logged into the CV builder app.

Postconditions: The user's profile information is updated in the app.

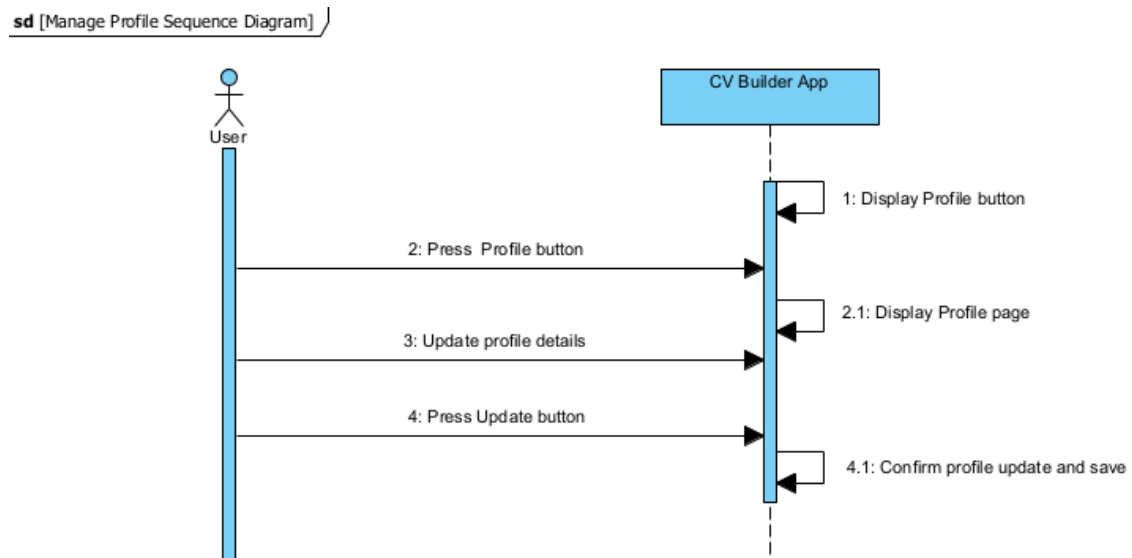


Figure 3.7: Manage Profile Sequence Diagram

Logout:

Actors: User

Description: The user can log out of the CV builder app, ending their session.

Preconditions: The user is logged into the CV builder app.

Postconditions: The user is logged out and cannot access functionalities that require authentication until they log in again.

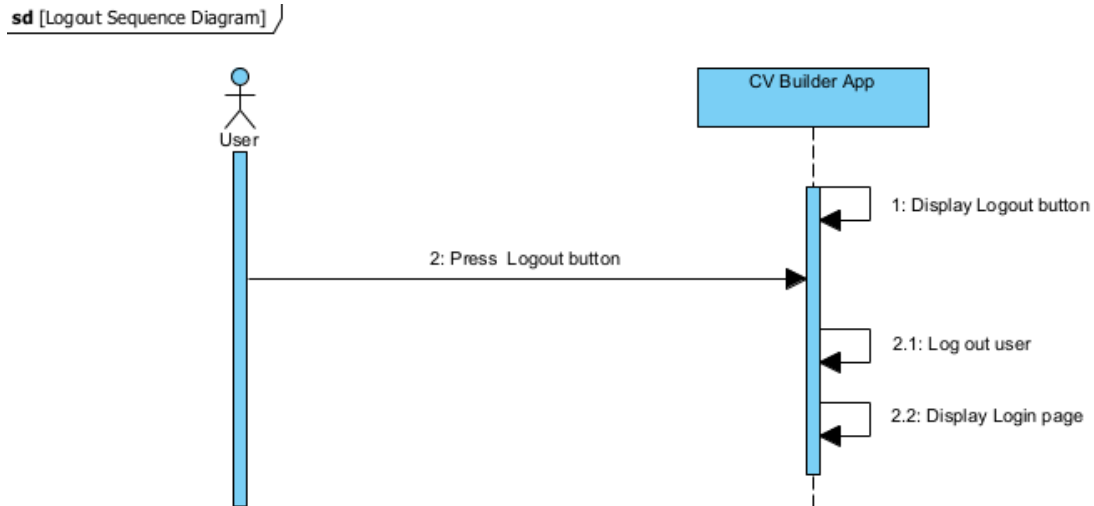


Figure 3.8: Logout Sequence Diagram

3.2 Application Graphical User Interface (GUI)

3.2.1 Introduction

"A user interface, in simple words, is the set of elements on a device or program that allow a user to interact with it."[6]

The graphic interface of the application aims to be simple and friendly for all types of users. The goal is to guide the user in such a way that he can finish his work as quickly as possible without wasting precious time.

3.2.2 List of Pages

In this section, an overview of the pages that make up the application is provided. Each page has a purpose and presents a series of elements that facilitate user interaction with the application. The pages are:

- **Hero Page:** This page is the first page that is displayed whenever someone accesses the web app. It presents a brief description of the application and presents buttons with links to the Register and Login pages. This is the only page that can be accessed without having a user account created on the application.

- **Login Page:** The Login page allows users who have created an account to log into the application using the email address and the unique password that they provided when the account was created.
- **Register Page:** The register page is where new users can go to create an account by providing their email address, name and password.
- **Home Page:** The homepage is very similar to the Hero Page, the difference being that the home page presents a broader description of the application. Also, from the Header you can interact with the other pages of the application. The header is present on all private pages.
- **Personal Information:** The Personal Information page is probably the most important page of the application. Here, the user enters and saves personal data such as an address, phone number, and links to social profiles, but also data related to academic background, work experience, skills, and personal projects. All this information is then used to create the CV.
- **Resume Pages:** There are 4 pages with CV templates. Their functionalities are similar, having a download button that the user can use to download a PDF copy of the CV. However, the design is substantially different between them, each having a different field of interest: Law, Computer Science, Sport, and Art.
- **Profile Page:** This page contains the identification data of the user. If he wishes, they can be modified once he presses the Update button.
- **Portfolio Page:** The implementation of the portfolio page is not yet finished. This page would act as a folder for previous user-created CVs. From here, he could edit, delete and download saved CV versions.

3.2.3 Page Elements and Analysis

- **Hero Page:**
 - **Elements:**
 - * **Brief Description:** It presents a brief introduction of the application
 - * **Register Button:** Link to the registration page
 - * **Login Button:** Link to the registration page
 - **Analysis:** The hero page is essential to attract new users. This page should have an engaging design and interface, enough to entice users to register or log in.
- **Login Page:**
 - **Elements:**
 - * **Email Address Field:** Where users enter their email address
 - * **Password Field:** Where users enter their password

- * **Login Button:** Submit the login credentials
- **Analysis:** The login page should have a simple and clear design, besides ensuring the confidentiality of the entered data. It is also very important to display error messages in case the information entered by the user is wrong.
- **Register Page:**
 - **Elements:**
 - * **Name Field:** Where users enter their name
 - * **Email Address Field:** Where users enter their email address
 - * **Password Field:** Where users enter their password
 - * **Confirm Password Field:** Where users enter their password again
 - * **Register Button:** Creates the account
 - **Analysis:** The registration page must keep all the design and security features of the login page. In addition, it is very useful to contain 2 fields for the password, in this way the number of accounts lost due to entering a wrong password during registration is considerably reduced.
- **Home Page:**
 - **Elements:**
 - * **Detailed Description:** A more in-depth overview of the application
 - * **Header:** Contains navigation links to the other pages of the application
 - **Analysis:** The home page should be informative and provide easy navigation to the other pages of the application.
- **Personal Information Page:**
 - **Elements:**
 - * **Data Entry Fields:** Fields to enter various personal information
 - * **Save Buttons:** To save the information provided in the above-mentioned fields
 - * **Header:** Contains navigation links to the other pages of the application
 - **Analysis:** This page is essential for creating a CV. It should handle user data securely, be well-organized, and be user-friendly.
- **Resume Pages:**
 - **Elements:**

- * **CV Templates:** Different designs for different fields (Art, CS, Law, Sport)
 - * **Download Button:** Download the selected CV in PDF version.
 - * **Header:** Contains navigation links to the other pages of the application
 - **Analysis:** These pages ought to provide attractive templates appropriate for the subject matter. The download procedure needs to be quick and error-free.
- **Profile Page:**
 - **Elements:**
 - * **Name Field:** Where users enter their modified name
 - * **Email Address Field:** Where users enter their modified email address
 - * **Password Field:** Where users change their password
 - * **Confirm Password Field:** Where users confirm their new password
 - * **Update Button:** Updates user data
 - * **Header:** Contains navigation links to the other pages of the application
 - **Analysis:** The profile page should be secure and accessible only if the user is logged in. It should provide an easy way to update the registration credentials.

3.2.4 Page Flow and Navigation

- **Diagram:**

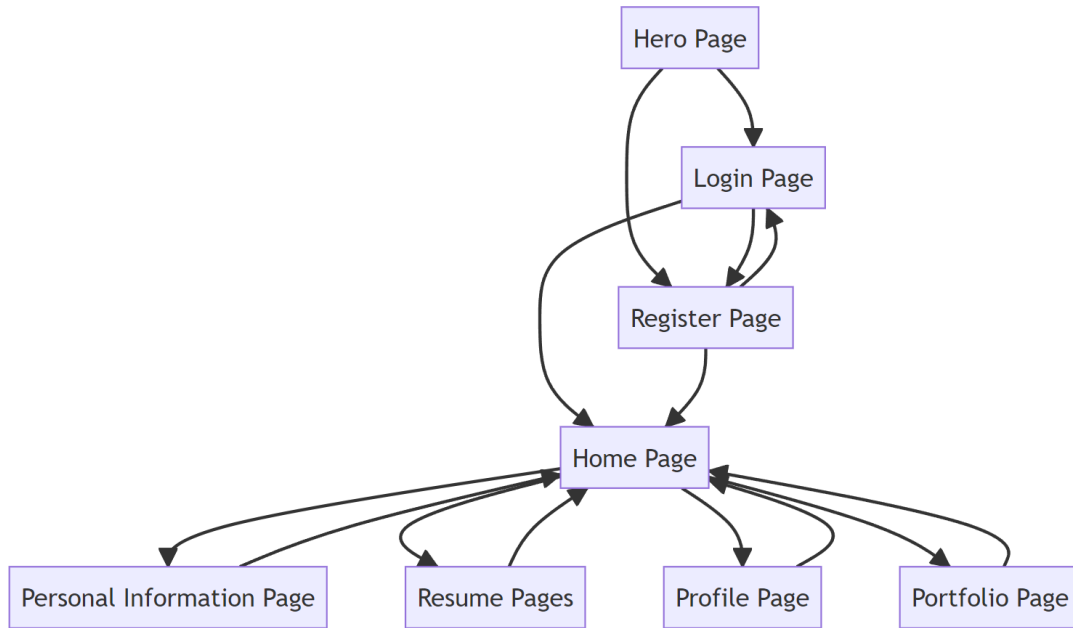


Figure 3.9: Flow between the pages inside the App

The diagram above shows the flow between the pages of the application. Arrows indicate possible routes from one page to another, and pages are represented by rectangular boxes.

- **User Navigation Patterns:**

- **Sequential Navigation:** Users often begin on the Home Page, move on to the Register or Login Page, and eventually arrive at the Dashboard Page, where they can access numerous functions.
- **Direct Navigation:** More experienced users can go straight to a particular page, such as the Personal Information Page, to edit their information.
- **Exploratory Navigation:** Before selecting a template, new users can browse a variety of options on the Resume Pages.

3.2.5 Screenshots

In this section, screenshots of the application pages will be presented.

The Hero Page from Figure 3.10 presents a clean and modern design, using bright colors to create a welcoming atmosphere. The call-to-action buttons are centrally positioned, being easy to spot.

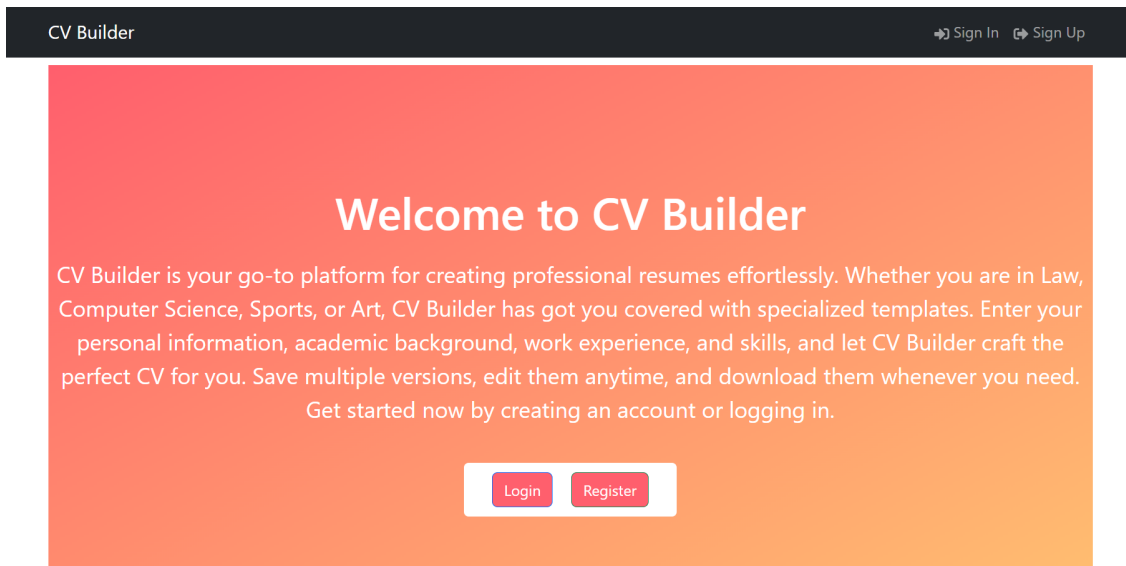


Figure 3.10: Hero Page

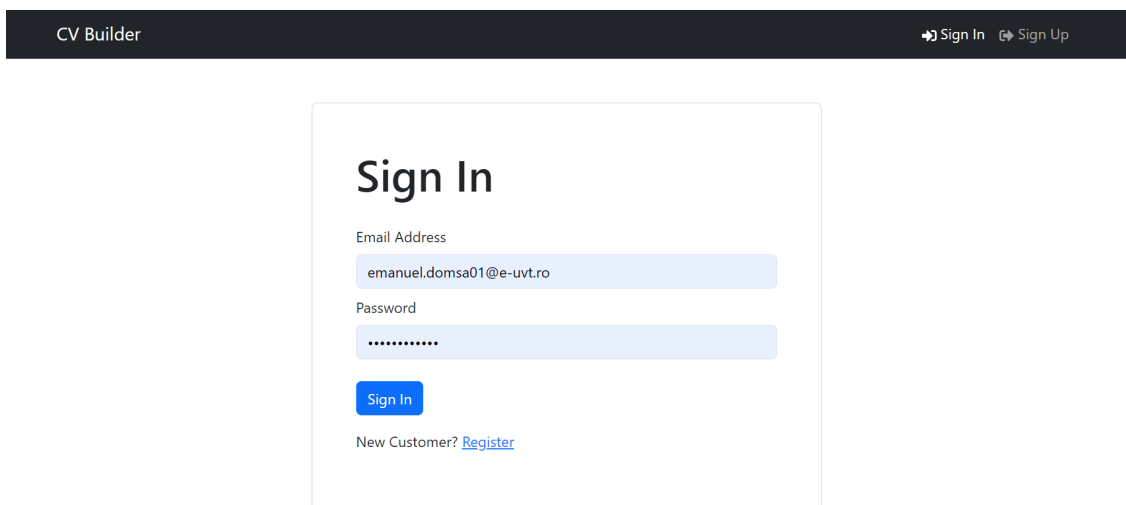
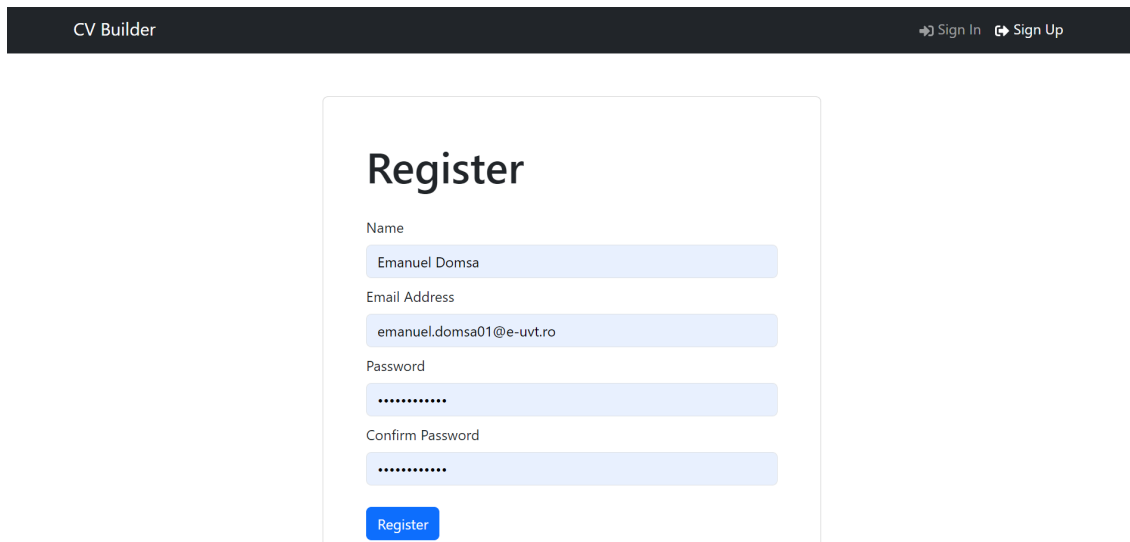


Figure 3.11: Login Page

In Figure 3.11 and Figure 3.12 the Login Page and Register Page are displayed. They have a simple and unencumbered design. The input fields and buttons are visible and well-sized.



The screenshot shows the 'Register' page of the CV Builder application. At the top, a dark header bar contains the text 'CV Builder' on the left and 'Sign In' and 'Sign Up' links on the right. The main content area is a light gray box with the title 'Register' in bold. Below the title are four input fields: 'Name' (containing 'Emanuel Domsa'), 'Email Address' (containing 'emanuel.domsa01@e-uvt.ro'), 'Password' (masked with dots), and 'Confirm Password' (masked with dots). A blue 'Register' button is positioned at the bottom of the form.

Figure 3.12: Register Page

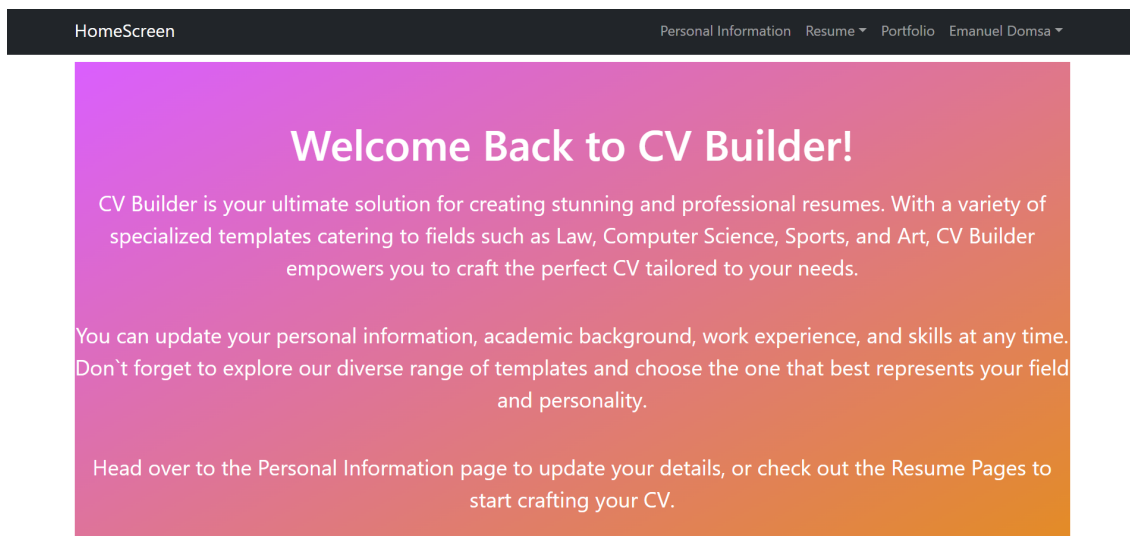


Figure 3.13: Home Page

The Home Page is shown in Figure 3.13. The design of the page is quite similar to the Hero Page in Figure 3.10, but with a slightly different color palette and without the call-to-action buttons since the user is already logged in. The text is easy to read and the information is to the point, providing a quick explanation of the purpose and operation of the app.

The screenshot shows a web application interface with a dark header. The header contains 'HomeScreen' on the left and navigation links 'Personal Information', 'Resume', 'Portfolio', and 'Emanuel Domsa' on the right. Below the header is a banner image of a laptop keyboard. The main content area has a profile section on the left with a circular profile icon, 'Your Name', and input fields for 'City, Name', 'Your Position', and 'Describe yourself in one line'. To the right of the profile section are links for 'LinkedIn', 'GitHub', and 'Portfolio', and input fields for 'Email' and 'Contact Number'. Below these are four expandable sections: 'About', 'Experience', 'Education', and 'Skills', each with a plus icon. At the bottom, there is a dropdown menu labeled 'Select CV to Choose from'.

Figure 3.14: Personal Information Page

The Personal Information Page presented in Figure 3.14 has a structured layout for input fields, making it simple for users to update and enter data.

The screenshot shows a web application interface with a dark header. The header contains 'HomeScreen' on the left and navigation links 'Personal Information', 'Resume', 'Portfolio', and 'Emanuel Domsa' on the right. Below the header is a 'Download' button. The main content area is divided into two columns. The left column has a light orange background and contains a circular profile icon, 'Your Name', 'Describe yourself in one line', 'Your Position', 'City, Name', and 'Skills'. The right column has a light gray background and contains 'About Me', 'Experience', and 'Education'.

Figure 3.15: CV Template Page 1

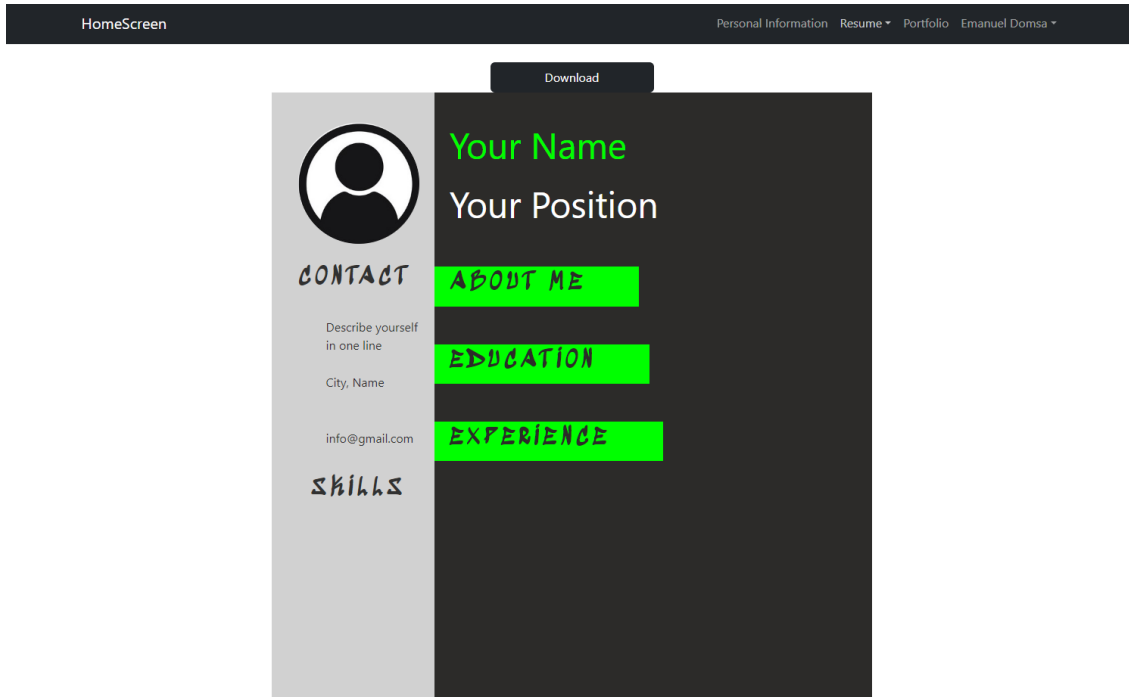


Figure 3.16: CV Template Page 2

The Resume Pages presented in Figure 3.15 and Figure 3.16 display 2 different templates with an emphasis on aesthetics. The download button is also very easy to spot.

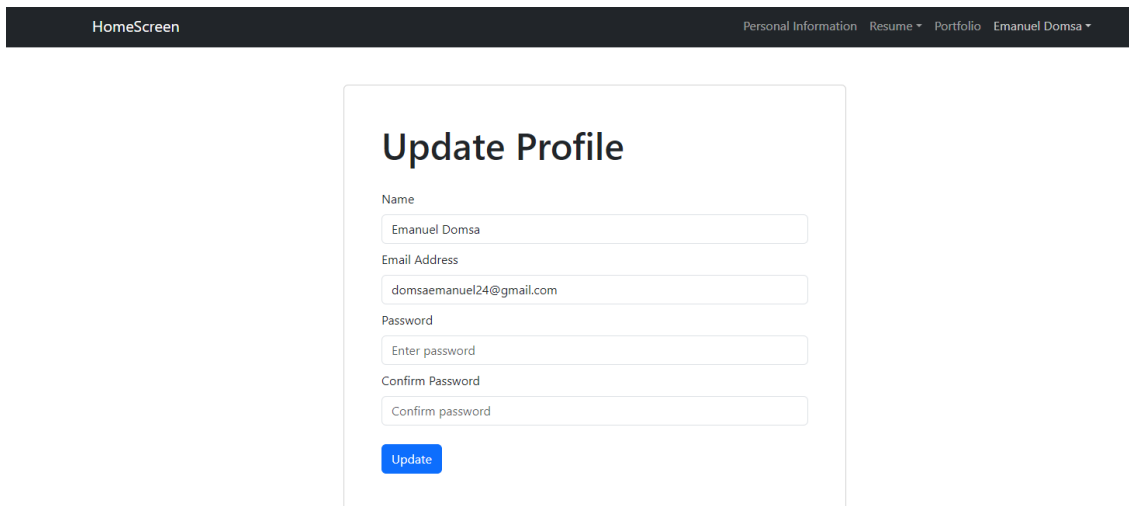


Figure 3.17: Profile Page

In Figure 3.17, the Profile Page is presented. The design is similar to the one from Figure 3.12, but the functionalities of the page are quite different.

3.2.6 Testing

In order to ensure the correct functionality of the CV builder application, a series of tests were conducted such as for creating a new user account, user authentication and profile update.

Starting with the creation of a new user account, three scenarios were tested. The first one represented correctly entering all the required data, which met the application's restrictions (such as password requirements). This resulted in a successful account creation and redirection to the main page.

The second scenario involved partially entering the data, (e.g. omitting some information) which led to the error messages being displayed. The third scenario involved entering a username or email that already existed in the database - or entering a different password at the verification input, this resulted in an alert indicating the duplication as seen in figure 3.18. .

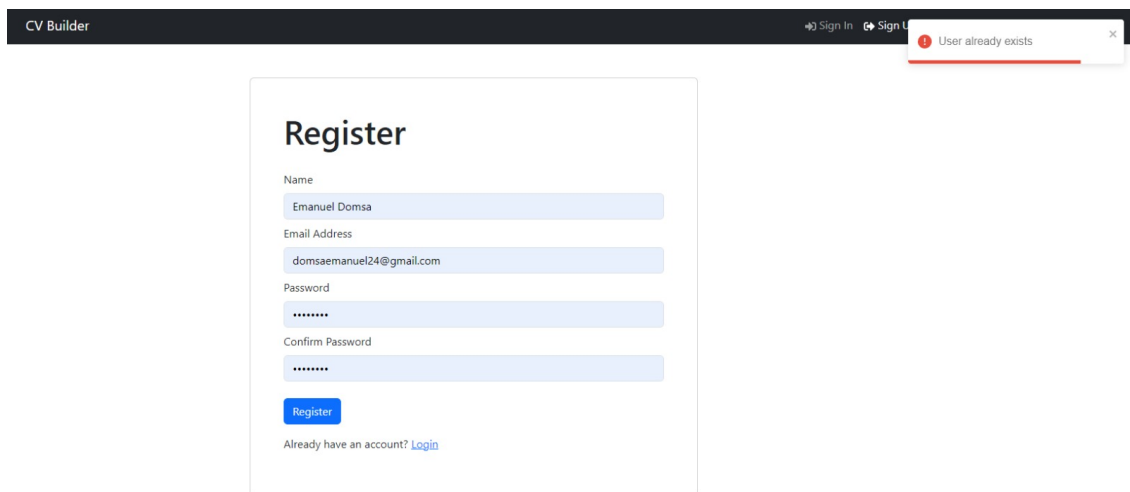


Figure 3.18: Register pop-up

To test the authentication functionality, two scenarios were considered. The first scenario involved a user successfully logging in by entering their username and password, and clicking the login button. The second scenario involved a user entering incorrect login information, resulting in the authentication process being halted and denying access to the rest of the application as figure 3.19 illustrates.

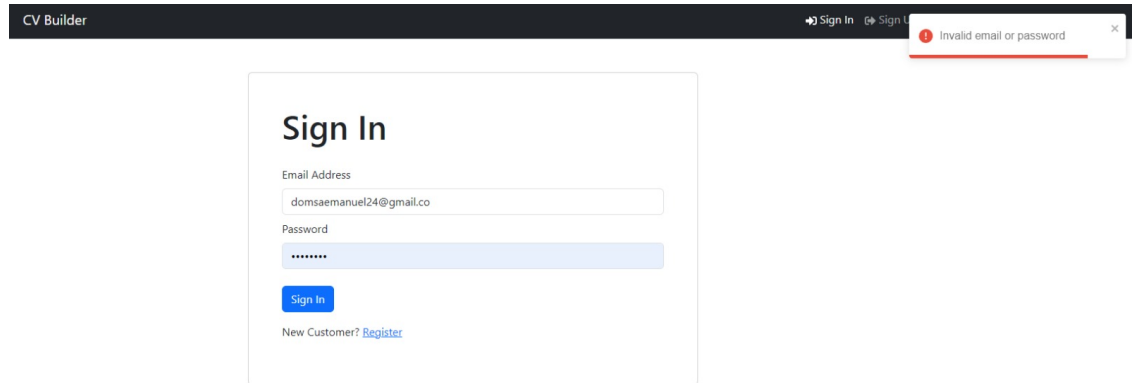


Figure 3.19: Sign in pop-up

For the profile update, the scenarios of the password not matching and the one where they do were taken into account - in the case that they do, the profile simply gets updated as seen in figure 3.20.

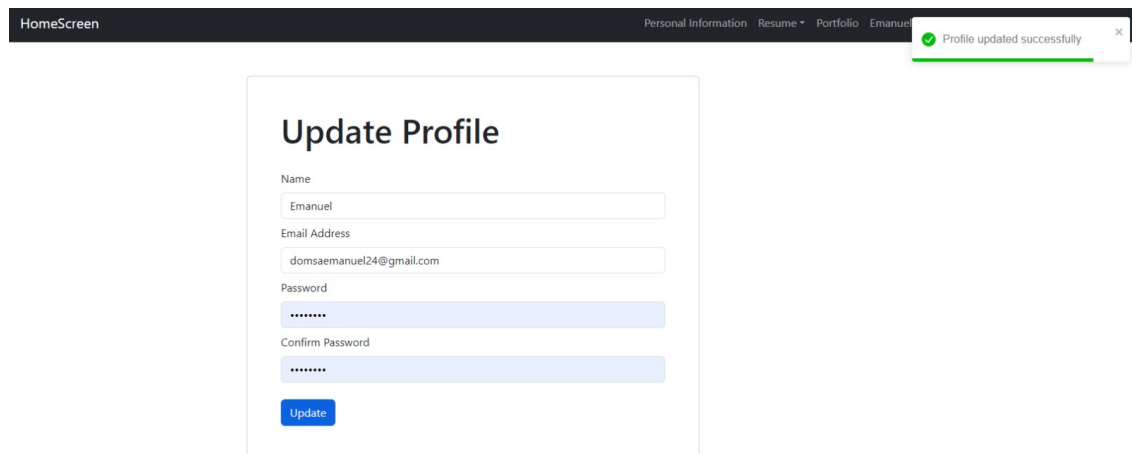


Figure 3.20: Update pop-up

Overall, these tests were conducted to verify the functionality, correctness, and user experience of the application, and to ensure that it meets the requirements and expectations of the users.

3.3 App Architecture

Three-tier architecture is a well-established software application architecture that organizes applications into three logical and physical computing tiers: the presentation tier, or user interface; the application tier, where data is processed; and the data tier, where the data associated with the application is stored and managed.[16]

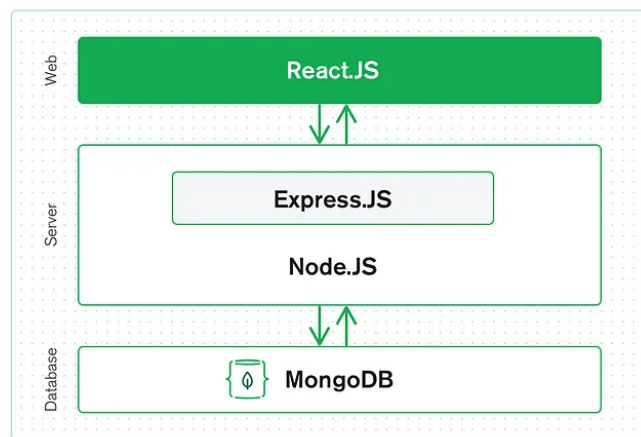


Figure 3.21: Three-tier architecture model[13]

3.3.1 Frontend

The frontend part of the application is divided into several modules. Each module is responsible for certain aspects of the application. The modular structure helps to better organize the code, making it easier to maintain and more flexible. Thus, if changes must be made to the code, only certain modules will be modified, without affecting the entire application. Also, under this structure, the code can be reused reducing duplication and redundancy.

A module is made up of a number of components with comparable functions and usage inside the application.

- **Components Module:** This module contains UI components that are used in different screens of the application. For example, the *Header* component appears in all application pages.
- **Small Components Module:** Similar to the previous module, this module also contains UI components. The difference is that the components inside this module are used only in some places by the components from the *Components* module.

- **Screens Module:** The Screens module contains all the components that form the pages of the application. All Screen components use one or more components from the Components module. Some of the components, such as *LoginScreen* and *RegisterScreen*, also use methods from the Slice module to communicate with the server to register or authenticate users.
- **State Module:** This module is responsible for managing the general state of the application, as it handles the data displayed and used by the UI components. For this Redux is used, it includes Redux store, actions and reducers.
- **Slice Module:** This Module has 3 components, *authSlice* is used to manage the authentication state of the app, *userApiSlice* is used to make API calls for user authentication and *apiSlice*, which is the base configuration for making API calls and is extended by *userApiSlice* to add specific endpoints for user management
- **Styles Module:** This module contains all the CSS files used to style some parts of the app.
- **Routing Module:** This module is formed from the *main* component and manages the routing of the application using React Router. It defines the paths and associates them with respective screen components.

3.3.2 Backend

Similar to the frontend part, the backend is also divided into several modules. The architecture for the backend is designed to be able to support the different functionalities of the frontend. It is built around a server that receives requests from the frontend, interacts with the database when it is necessary to add or extract data, and responds to requests from the frontend.

- **Config Module:** This module is responsible for establishing a connection to the MongoDB database using the Mongoose library. It ensures that the application and the database can communicate with one another.
- **Controllers Module:** It contains the logic to handle user-related operations. Such operations are authentication, registration, requesting and updating user profiles. It interacts with the Config model to perform create, read, update and delete (CRUD) operations on the user data in the database and utilizes JWT for handling authentication.
- **Middleware Module:** This module has 2 components, *authMiddleware* which protects certain routes by checking the validity of the JWT, and *errorMiddleware* whose role is to handle error throughout the application.
- **Models Module:** This module defines the schema for the user in the database, using fields such as name, email and password. Besides that, it also has methods by which it hashes passwords and methods by which it compares passwords once they are hashed.
- **Routes Module:** This module serves as the orchestrator for routing user-related requests within the Express.js application. It meticulously organizes

the endpoints associated with user operations and maps them to the appropriate controller functions. Additionally, it incorporates an authentication layer to certain routes, ensuring that only authenticated users have access. This structured approach enhances maintainability and security within the application.

- **Utils Module:** It has a single task, to generate a JSON Web Token for authentication. However, this module is crucial for the security of the application. The token that it generates can be used to verify the identity of the users when they are using the app.
- **Server Module:** It contains a script that initializes the Express.js server, connects to the database, configures middleware for request processing, sets up API routes, handles static files in production, and implements custom error handling. It serves as the backbone of the backend, orchestrating various components for a seamless operation.

In Figure 3.22 we can see a graphical representation of the backend and frontend. There is a clear structure on the frontend, communication with the backend is done through the Slice module. On the backend, it is very easy to see that the Server module has access to all the specific routes, and the database and it's at the receiving end of API requests from the frontend.

pkg (Package Diagram FR 86)

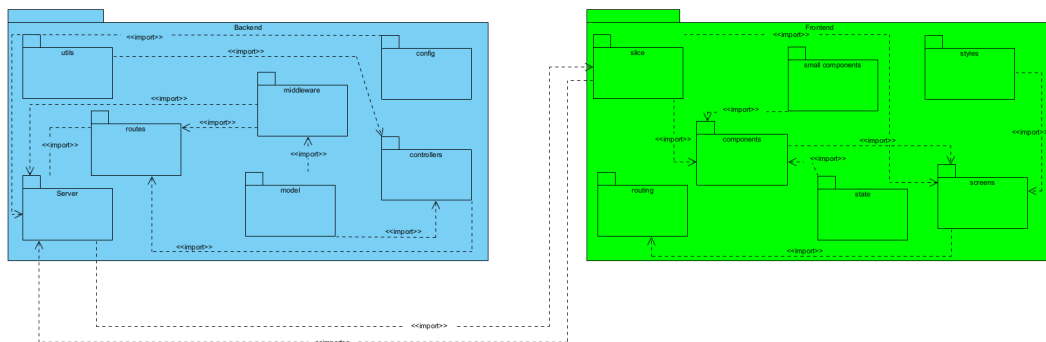
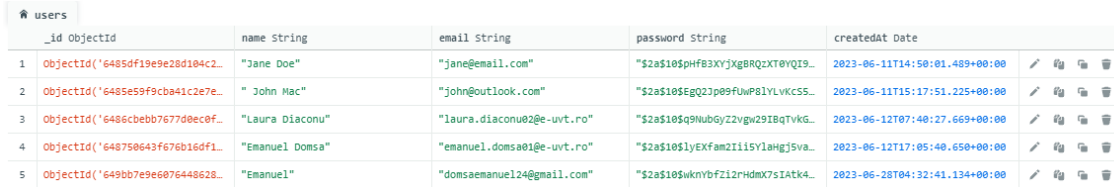


Figure 3.22: Frontend and Backend Structure

3.3.3 Database

In regards to the building process of this app, in order to store the data, a NoSQL database was used, due to the fact that not only is it easier to use, but also more beginner-friendly.



	_id ObjectId	name String	email String	password String	createdAt Date	
1	ObjectId('6485df19e9e28d104c2...')	"Jane Doe"	"jane@email.com"	"\$2a\$10\$phF83XyJXg8RQzXT0YQI9..."	2023-06-11T14:50:01.489+00:00	  
2	ObjectId('6485e59f9c8a41c2e7e...')	"John Mac"	"john@outlook.com"	"\$2a\$10\$EgQ2Jp09fUwP81VLkc5S..."	2023-06-11T15:17:51.225+00:00	  
3	ObjectId('6486cbebb7677d0ec8f...')	"Laura Diaconu"	"laura.diaconu82@e-uvf.ro"	"\$2a\$10\$qNubGyZ2vgw2918qTvkG..."	2023-06-12T07:40:27.669+00:00	  
4	ObjectId('648750643f676b16df1...')	"Emanuel Domsa"	"emanuel.domsa01@e-uvf.ro"	"\$2a\$10\$lyEXfam21i15YlaHgJ5va..."	2023-06-12T17:05:40.650+00:00	  
5	ObjectId('649bb7e9e6076448628...')	"Emanuel"	"domsaemanuel24@gmail.com"	"\$2a\$10\$wknYbFz12rHdmX7sIAtk4..."	2023-06-28T04:32:41.134+00:00	  

Figure 3.23: Users Table

In the Figure 3.23 is an example of how the user's data (email address, password) is stored inside the database. Additionally, there are another 2 columns, one with the user ID, whereas, as a result of a callback upon editing any information, the second column - which initially is set to the creation of the user account.

3.3.4 Used technologies

The MERN stack was used in conjunction with the JSON Web Token (JWT). In regards to the frontend, the main development language used was React, a library of JavaScript, with an occasional needed usage of Redux and ReactDOM; the latter being used to achieve an increase in performance.

- **MongoDB:** MongoDB is a NoSQL database that was used to store and manage data for the application. It was chosen for its flexibility and scalability. In the application, MongoDB stores user information, authentication tokens, and other relevant data.[14]
- **ExpressJs:** Express.js is a web application framework for Node.js. It was used to create the backend of the application. Express.js made it easy to define routes, handle HTTP requests, and integrate middleware for processing requests.[5]
- **ReactJs:** React.js is a JavaScript library for building user interfaces. It was used to create the frontend of the application. With React.js, the application's UI was built using reusable components, which made development more efficient and the user interface more responsive.[15]
- **NodeJs:** Node.js is a JavaScript runtime built on Chrome's V8 JavaScript engine. It was used to build the server-side of the application. With Node.js, the application could handle asynchronous operations and manage multiple connections efficiently.[10]
- **JSON Web Token (JWT):** JWT is a compact, URL-safe token format used for securely transmitting information between parties. In the application, JWT was used for authentication and authorization. When users log in, they

receive a JWT, which must be sent in the header of HTTP requests to access protected routes and resources.[9]

Chapter 4

Conclusions and Future Work

4.1 Conclusion

The purpose of this bachelor's thesis was to create a CV Builder web application. The requirements for this were clear, to have a welcoming and easy-to-use interface, and to present the user with clean and professional-looking resume templates from which to choose.

And that was accomplished, the application achieves all its original goals, offering an intuitive UI with reusable components. This facilitates the user experience, while also making updates and maintenance easier. There were some setbacks during the development process of this application, but they presented a great learning opportunity.

In conclusion, the application was developed successfully and it was an unique and complex experience.

4.2 Future Work

Although the application works according to the goals set at the beginning of the development process, there is always room for improvement and innovation. These are planned for future application development:

- **More CV Templates:** As of right now, the application offers only 4 templates. In order for the app to stay relevant and attract more users, a larger selection of templates is needed.
- **To import user data directly from the LinkedIn profile:** It is quite common nowadays for apps of this type to be able to extract personal data by only having access to your LinkedIn profile. It's not an easy task to implement such a feature, but it's a must and the challenge is up.
- **To have the option to create a Cover Letter:** Almost all the applications that were used as inspiration had such a feature implemented, so it is only natural for the **CV Builder App** to have one.

By implementing these features, the application will become more versatile and engaging, which will consequently attract more users.

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